Calling PerfDesign For Perf Boards, Strip Boards

PRIYA RAVINDRAN

A breadboard is like staple food to every electronics enthusiast. This is the first thing you would turn to when there is a circuit to design. Converting the lines and curves into signals on the oscilloscope or figures on a digital display is what you as a designer would put your entire efforts into.

Everything works and then comes the most important piece—the board. It is the board that decides the fate of the circuit. But what if you are working with perf and strip boards? How do you go about laying out the circuit? PerfDesign is here to help you!

PerfDesign is a tool for quickly designing or sketching projects on perf and strip boards. Get your Windows laptop, fire up your system and install the package. In a matter of minutes, you will be good to go. But do you even need software to create a design on a simple perf board?

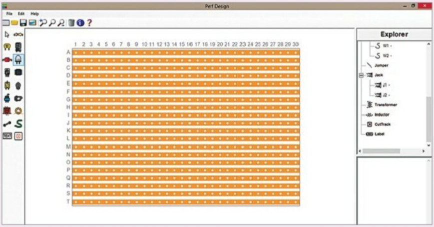

Fig. 1: PerfDesign layout with a strip board

Why PerfDesign

When I worked on my first big project, I started off by creating a working prototype on a breadboard. My professor then told me to replicate the same on a general-purpose PCB—a plain flat board with rows and columns of holes. Translating to the new board, which I later got to know was indeed the perf board, resulted in more stability, lesser noise, an easy-to-handle board and basically better results.

I remember sitting down with a pencil and paper in hand, tracing out graph-like checked squares and trying to put down components on it to scale. I began by drawing a resistor as big as the original one at hand, counting the number of dots before placing the second resistor to make space for node B, and so it went.

After many a drawing and erasing, I managed to get my final circuit to look the way I wanted—component placement, solder, wires and spacing between everything, all taken care of. I wanted the circuit to be neat, after all!

The main problem with designing like this, apart from the sheer number of iterations, is for a layout that involves a lot of cross-crossing; the drawing needs to be that much more accurate!

How much more difficult would it be for creating paths for connecting the ports for I/O devices, or maybe revamping the circuit to accommodate a new design? Enter, PerfDesign!

The crosschecks

PerfDesign is a software that lets you design perf or strip board layouts. This is an easy-to-use tool that helps you quickly put together a layout on a custom PCB. Begin by laying out the components, make way for nodes by moving everything around and finish by connecting the design via traces. I find it easier to use and debug by keeping the layout as close to the schematic as possible.

When you start a new project on PerfDesign, you either choose the perf board or go with the strip board model. How do you choose, though?

The perf board is a set of holes arranged in neat rows and columns. The strip board is exactly the same, except that the holes along each row are connected via copper traces. The former lets you design everything just the way you want, while the latter is easier to work on if all you want to do is replicate the design on

A look at PerfDesign

- Software written in C#
- Released under GNU GPL licence
- Requires Microsoft NET framework 4 or higher
- Authored by AndyBandy
- Works best on Windows 7
- PerfDesign 1.0.0.0 is the latest version